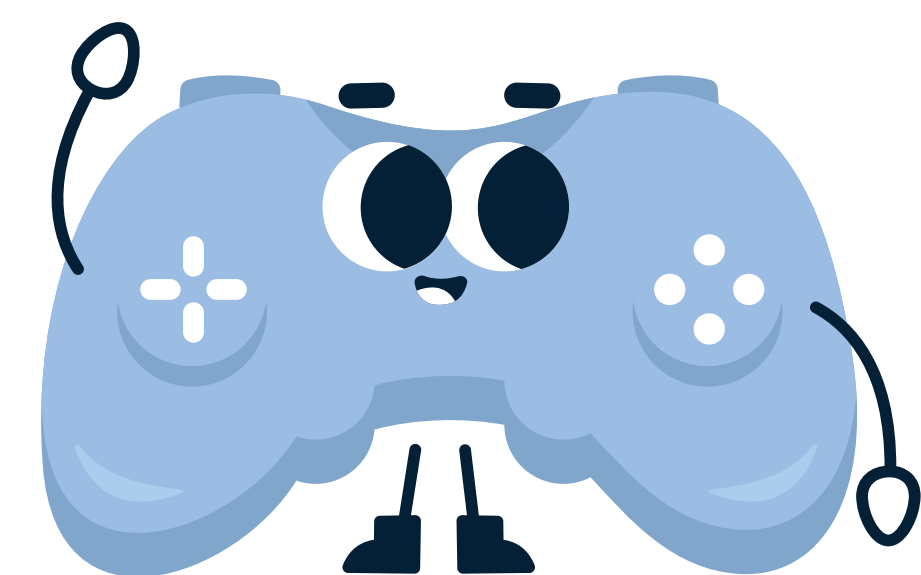


PREDICTIVE PLAY: DUAL-ENGINE COLLABORATIVE FILTERING



Authors

Maria Mahapa

Panashe Gomora

Johannes January



Affiliations

Quinnipiac University

BAN 628 Data Mining

Academic Research Poster

Abstract

This research uses user-based and item-based collaborative filtering to tailor board game recommendations from the BoardGameGeek dataset, featuring 1.94 million ratings. A Bayesian Gold Standard weighting method filters noise, showcasing popular games like Gloomhaven and war-strategy related, thematic niche games for user 205614.

01 PROJECT INFRASTRUCTURE & SCALE

21,616

UNIQUE GAMES

261K

USERS

1.94M

RATINGS

- Sparse Matrix: Memory reduced to 23 MB for crash-free processing.
- Digital ID Mapping: Unique ID for each user allows quick lookup.
- Focus User: ID 205614 as the primary test case.
- Modern Scope: Catalog includes games published since 1900.

02 BUSINESS QUESTIONS

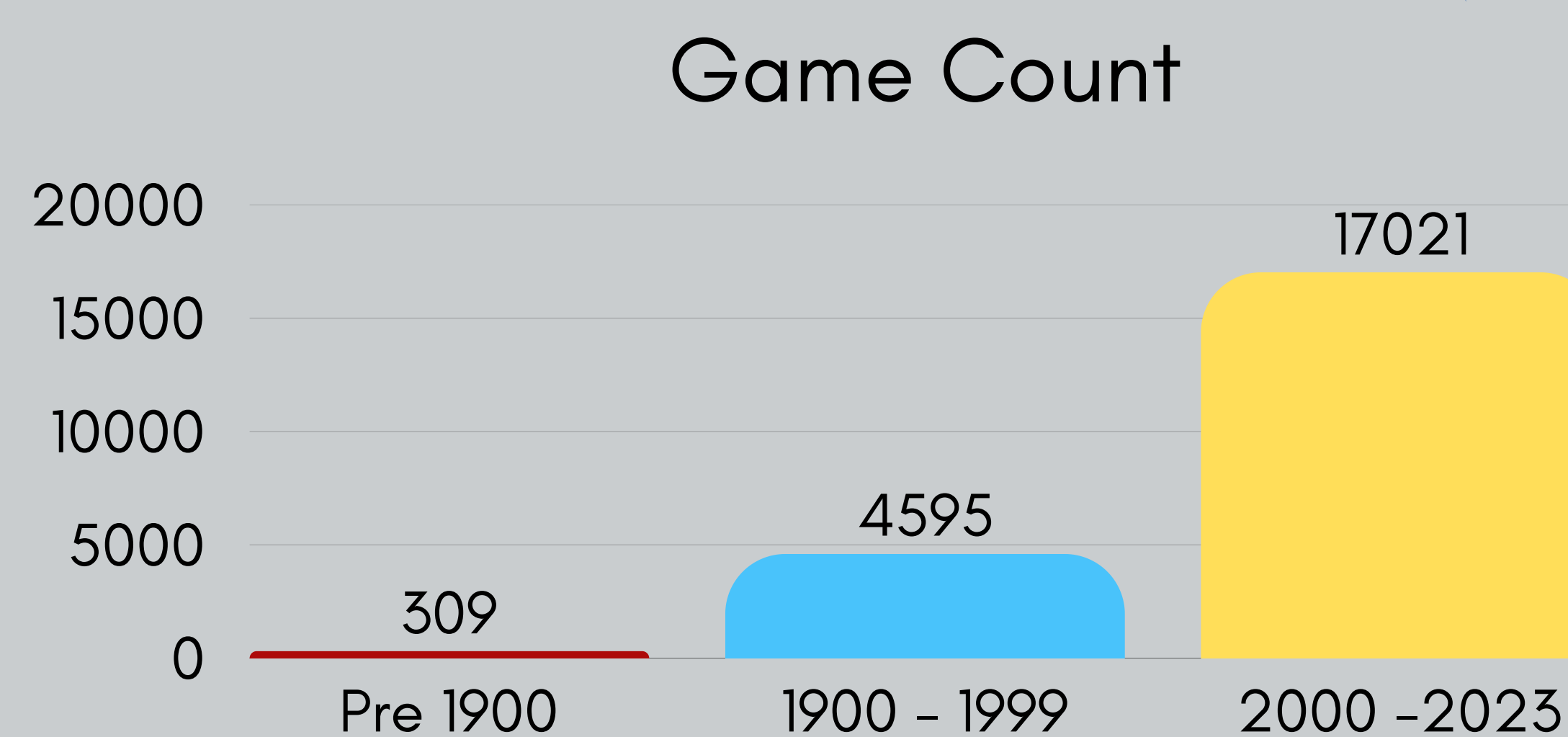
BQ1 - What is the user's next game? based on test twins.

BQ2 - If a user enjoys Game X, what other games might they like?

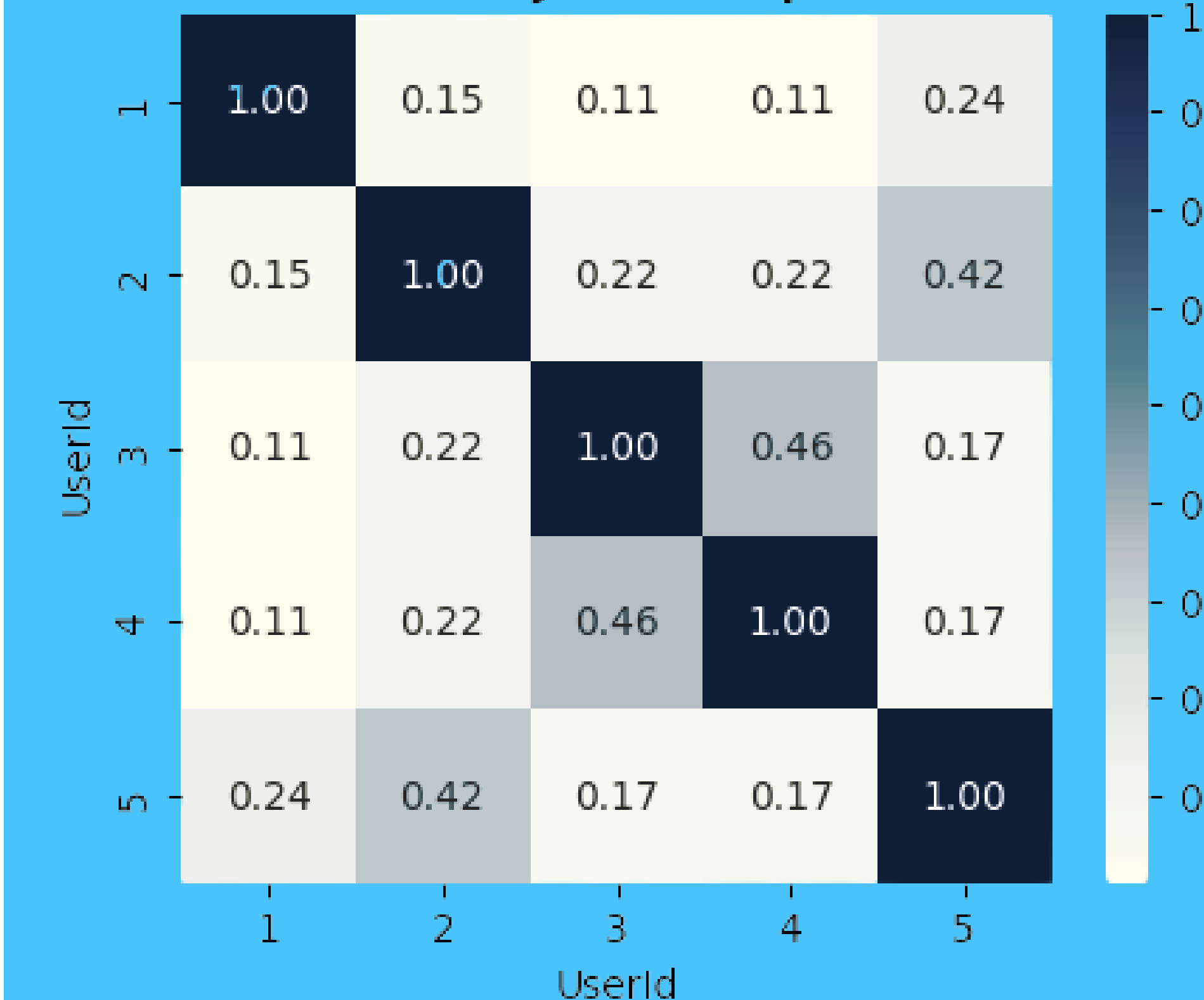
03 METHODOLOGY

- User-Item Collaborative Filtering
- Item-User Collaborative Filtering
- Bayesian weighting

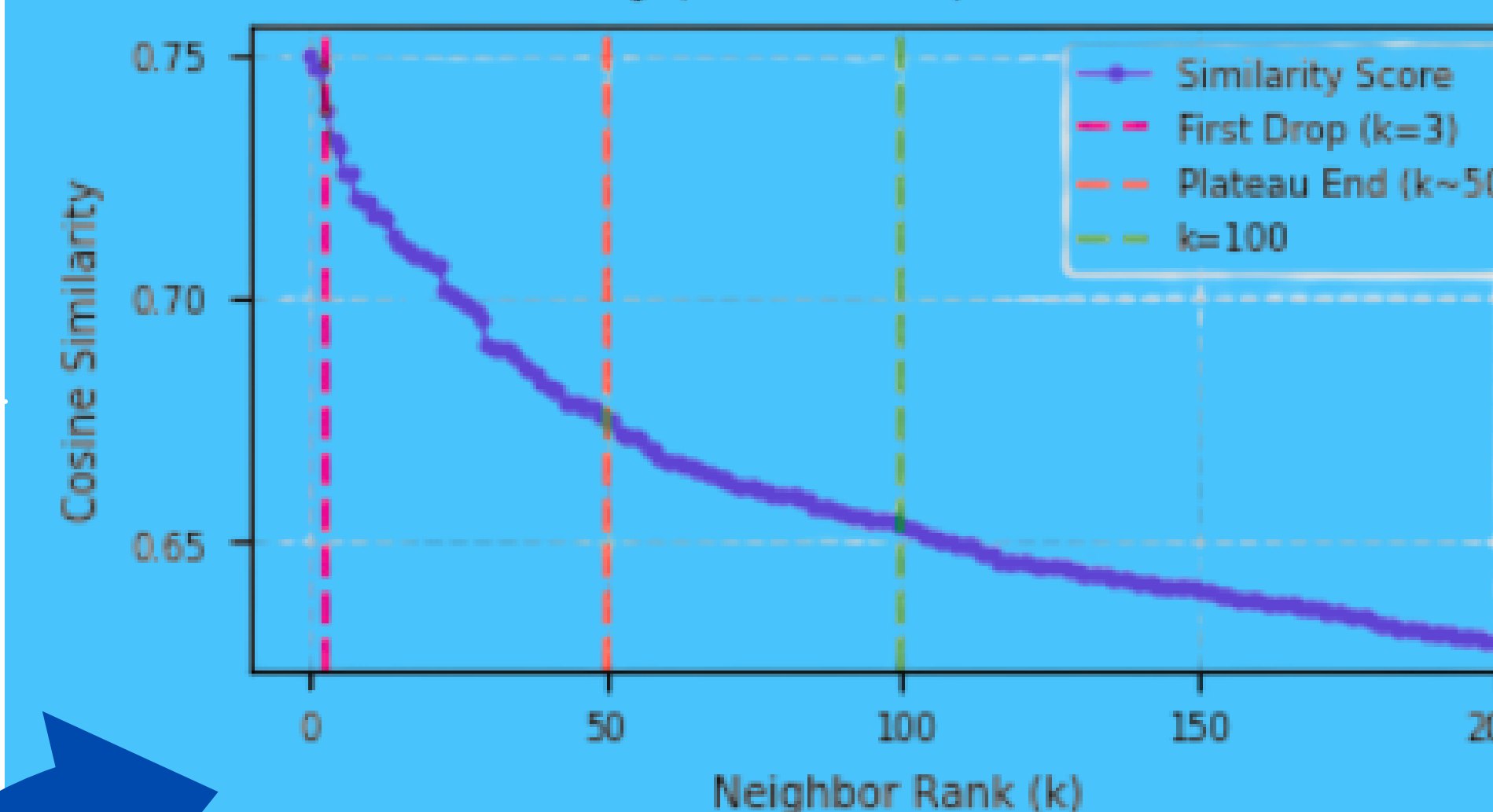
04 VISUALISATIONS



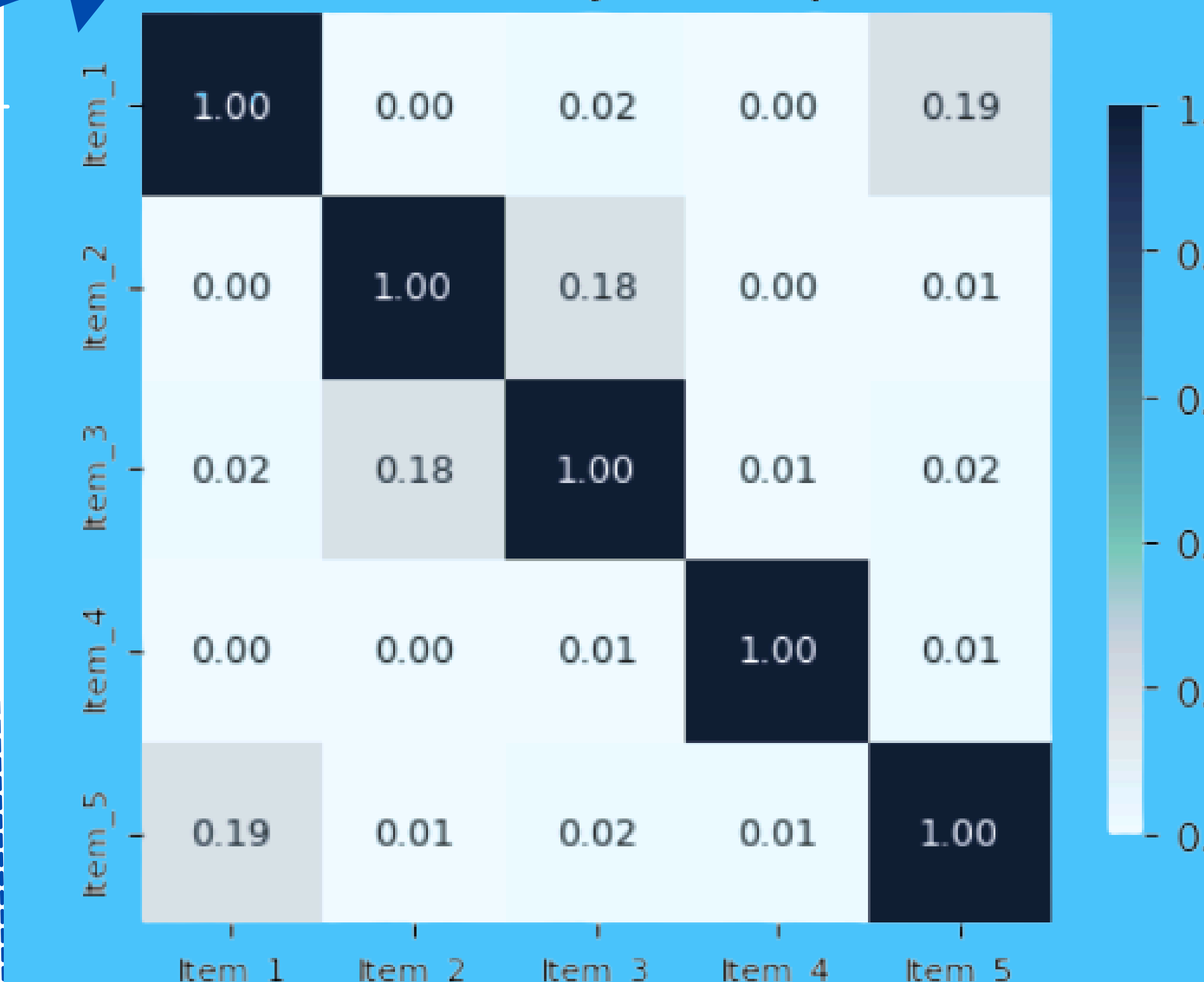
User-User Similarity Heatmap (First 5 Users)



Similarity (k=0 to 200) for User 205614



Item-Item Similarity Heatmap (First 5)



U-B-F GAME RECOMMENDATIONS

#Pop	#Wtd	Game	Cat	Yr	Nbrs	Avg	W.sc	Move
—	1	Gloomhaven	Them	2017	9	7.49	6.91	NEW
5	2	Scythe	Stra	2016	35	7.00	6.86	▲3
8	3	Castles of Burgundy	Stra	2011	27	6.99	6.81	▲5
7	4	Agricola	Stra	2007	29	6.70	6.58	▲3
1	5	Pandemic	Stra	2008	47	6.58	6.51	▼4
—	6	Pandemic Legacy S1	Them	2015	11	6.77	6.50	NEW
3	7	Codenames	Part	2015	41	6.55	6.48	▼4
10	8	T2R: Europe	Fam	2005	25	6.57	6.46	▲2
—	9	Power Grid	Stra	2004	23	6.52	6.41	NEW
—	10	Pablo	N/A	2017	1	9.00	6.40	NEW
2	—	Carcassonne	Fam	2000	46	6.39	—	DROP
4	—	Catan	Strat	1995	39	5.87	—	DROP
9	—	Dixit	Part	2008	25	5.81	—	DROP
6	—	Ticket to Ride	Fam	2004	33	6.25	—	DROP

I-B-F GAME RECOMMENDATIONS

#Wtd	#Raw	Game	Cat	Yr	#Rtn	W.Sm	R.Sm	Move
1	1	Santa Cruz 1797	War	2017	136	0.14	0.19	—
2	3	Crusade & Revolution..	War	2013	472	0.14	0.15	▲1
3	2	Coral Sea: ..	War	2010	116	0.11	0.15	▼1
4	6	Plus Ultra: ...	Strat	2016	166	0.11	0.14	▲2
5	5	Roads to Stalingrad:	War	2009	138	0.11	0.14	—
6	7	Triumph & Tragedy: ...	War	2015	1876	0.11	0.11	▲1
7	9	MIL (1049)	Stra	2011	687	0.10	0.11	▲2
8	—	1911 Amundsen vs Scott	Them	2011	1022	0.10	—	NEW
9	—	Polis:...	Strat	2012	2205	0.09	—	NEW
9	10	ONUS! Rome Vs Carthage	War	2014	293	0.09	0.10	—
—	4	Punic Island: CC Volume III	War	2011	56	—	0.15	DROP
—	8	Reconquista	War	2018	70	—	0.11	DROP

05 ANALYSIS U-B-F

- Era: Modern Era (2011-2017)
- Complexity: High (strategy games)
- Recommendation: Discover top-rated modern masterpieces

ANALYSIS I-B-F

- Era: Modern Era (2009-2018)
- Complexity: Med - Heavy
- Recommendation: Discover top-rated modern strategy games & niche focused suggestions ie War

06 FINDINGS

- User 205614's Taste Twins favor complex modern hits.
- Mediterranean Empires (Strategy) introduces more modern games and niche war / thematic games
- Features a Bayesian Tag of War effect.

LIMITATIONS

- Cold Start Problem.
- Popularity Bias:
- Sparsity: Users rate only a few of the 21,000+ games, missing hidden connections / 19%.
- Subjectivity: Ratings may reflect personal experiences rather than game quality.